

# PAVAN KUMAR SADASHIV

## Backend Software Engineer | Distributed Systems & High-Performance Runtimes

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## PROFESSIONAL SUMMARY

High-velocity Backend Software Engineer and Systems Architect with specialized expertise in high-concurrency API services, asynchronous background worker pipelines, resilient database schema design (PostgreSQL/SQLite), and low-level performance optimization. Active member of the **GitLab Community Contributor Program** with direct hands-on experience navigating community forks, MR review triages, and automated CI/CD pipelines. Founder & Managing Director of HRL International Private Limited, delivering audited software architectures with complete end-to-end feature ownership and disciplined AI-augmented development velocity.

## CORE TECHNICAL COMPETENCIES

- **Backend & Systems:** Python (FastAPI, Asyncio, Multiprocessing), Go (Concurrency, Goroutines), Node.js (Zero-Framework, REST APIs), C++20, Rust, Microservices Architecture.
- **Databases & Storage:** PostgreSQL (Schema Evolution, Indexing, Query Optimization), SQLite 3 (ACID Transactions), Paged Virtual Allocation, In-Memory Caching.
- **DevOps & CI/CD:** GitLab CI/CD (.gitlab-ci.yml), GitHub Actions, Docker, Linux/POSIX System Administration, Automated Unit & Integration Testing.
- **Algorithms & Architecture:** Combinatorial Optimization (CSP Backtracking), Sliding Window Rate Limiting, Modular Monolith Architecture, Bounded Contexts.
- **AI Fluency & Tooling:** GitLab Duo (Code Suggestions, Chat), LLM Inference Pipelines, Speculative Decoding, Google Antigravity SDK, Critical Code Output Validation.

## FLAGSHIP BACKEND & DISTRIBUTED SYSTEMS PROJECTS

## GitLab Community Contributor — Core & Satellite Ecosystem

Ecosystem: GitLab SaaS & Community Forks, GitLab Duo, CI/CD Components | [gitlab.com/hrlpavan](https://gitlab.com/hrlpavan)

- Onboarded to the official **GitLab Community Contributor Program** (Work Item #5499) with active development across community forks.
- Focused on satellite backend and tooling projects: **AI Gateway** (Python/FastAPI), **Code Parser & Knowledge Graph** (Rust), and **VS Code Extension**.
- Integrated live dynamic SVG contributor telemetry banners into public portfolio profiles, tracking real-time GitLab Contributor Platform metrics.

**Nirman-Drishti (■■■■■■■■-■■■■■■■) — Autonomous Forensic Audit & Backend Analytics Engine**

Technologies: Python, FastAPI, Asyncio, PyArrow, Multi-Agent Orchestration | [github.com/hrlpavan/nirman-drishiti](https://github.com/hrlpavan/nirman-drishiti)

- Architected a high-throughput backend data pipeline for public infrastructure and forensic auditing, ingesting multi-source telemetry data.
- Engineered asynchronous batch processing workflows utilizing PyArrow for zero-copy memory operations, reducing data ingestion overhead by 70%.
- Implemented defensive input sanitization, automated audit trails, and strict transaction logging to guarantee data immutability.

## InvigiMatrix — College Examination & Scheduling Management Backend

Technologies: JavaScript (ES6+), Native Node.js (Zero-Framework), SQLite 3, CSP Backtracking | [github.com/hr/pavan/exam-invigilation-dsa-app](https://github.com/hr/pavan/exam-invigilation-dsa-app)

- Built a zero-external-dependency native Node.js REST API server backed by SQLite with strict ACID-compliant transaction boundaries.
- Implemented an NP-hard **Constraint Satisfaction Problem (CSP) Backtracking solver** eliminating multi-department scheduling collisions.
- Designed sliding window token bucket rate limiters, modular service endpoints, and automated data validation guards preventing race conditions.

## RTX-LocalAI Runtime — High-Performance Systems & Execution Suite

Technologies: C++20, CUDA 12.6, Paged KV-Cache, Concurrency, CMake | [github.com/hrlpavan/nvidia-project-by-hrl](https://github.com/hrlpavan/nvidia-project-by-hrl)

- Engineered an on-device high-performance inference engine featuring virtual-to-physical block memory management (16 tokens/block).
- Achieved **0.17  $\mu$ s allocation latency** with **0.00% memory fragmentation**, featuring static compute graph replay (< 5  $\mu$ s launch latency).

## PROFESSIONAL EXPERIENCE & LEADERSHIP

**Founder & Managing Director / Chief Systems Architect**

*HRL International Private Limited | Mangaluru, India | 2026 – Present*

- Direct end-to-end software architecture, code quality, and delivery for enterprise AI, backend services, and distributed compute platforms.
  - Spearheaded statutory and technical compliance for **Startup India Seed Fund Scheme (SISFS)** and **Karnataka ELEVATE Grant 2026**.
  - Championed an asynchronous, remote-first engineering culture with documentation-driven specifications and CI/CD-enforced quality gates.
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## EDUCATION & CERTIFICATIONS

### **Bachelor of Engineering (B.E.) in Computer Science & Engineering (AI & ML)**

*Sahyadri College of Engineering & Management (SCEM), Mangaluru, Karnataka, India*

- Core: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Computing.
- **JPMorgan Chase & Co.:** Distributed Streaming & Software Engineering Simulation (Kafka, REST API Architecture).
- **Deloitte Australia:** Cybersecurity Defense & Threat Modeling Simulation (Perimeter Defense, Identity & Access).